

**FoodFlow AI**

CORNELL CBC HACKATHON 2026 · SOCIAL IMPACT

The Autonomous Food Rescue Engine

AI-powered surplus prediction and real-time redistribution — connecting institutions to food banks before the food ever hits the bin.

119M

lbs wasted daily in the US

47M

food-insecure Americans

\$2.2B

addressable market

380x

LTV:CAC (enterprise)

Claude tool_use

Cornell Dining Pilot

Feeding America Network

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— THE PROBLEM

Two crises exist side by **side** — and never meet.

119M

lbs of food wasted every single
day in the US

ReFED 2024 — 30–40% of US food
supply

47M

Americans are food-insecure,
including 13M children

USDA Economic Research Service,
2024

\$218B

in food value lost annually — 2%
of US GDP

EPA Food Waste Hierarchy, EPA.gov
2024

WHY SURPLUS NEVER REACHES FOOD BANKS

**The Timing Problem**

Surplus is known only at closing time — 8pm. Food banks need 2–4 hours lead time to dispatch volunteers. By the time a coordinator is called, it's too late.

**The Coordination Problem**

Today's "system" is phone tag. A dining hall manager calls a food bank coordinator who calls a volunteer driver who may or may not show up. No platform, no tracking, no reliability.

**The Data Problem**

No institution can prove how much food they rescued, when, or what impact it had. ESG reporting is done with spreadsheets and guesswork. Donors have no accountability layer.

THE BROKEN STATUS QUO — A REAL STORY

It's 8:47pm. Cornell Statler Hall has **80 lbs of pasta** left over from the banquet. The manager calls Maria at the Ithaca Food Bank. She has no driver available. It's too late to coordinate.

By 9:15pm, the pasta is in the bin. That same night, **400 families** in Ithaca went to bed without enough food.

This happens **every single day**.

40%

of food bank demand goes unmet in the US due to
supply unpredictability

Feeding America, 2024 State of Hunger Report

— THE SOLUTION

Predict. Match. Dispatch. Automatically.

1

Predict Surplus 2 Hours Early

FoodFlow integrates with POS and inventory systems. Our ML model analyzes order patterns, historical waste, and event data to predict surplus food — type, quantity, and availability time — 2 hours before closing.

 POS Integration

2

Claude AI Orchestrates the Match

A Claude tool_use agent runs 6 autonomous API calls: checks food bank capacity, queries volunteer availability, calculates optimal route, verifies Bill Emerson Act compliance, and issues a dispatch decision — in under 4 seconds.

 Claude API

3

Auto-Generate ESG Reports

Every rescue is logged: pounds diverted, meals served, CO₂ avoided, landfill fees saved. Monthly branded ESG reports are generated automatically — no spreadsheets, no data entry. Institutions get compliance data in one click.

 ESG Auto-Report

HOW CLAUDE'S AGENT WORKS — LIVE TOOL_USE LOOP

```
[FOODFLOW AGENT] Surplus detected: 78 lbs pasta
[TOOL: check_inventory] → confirmed 8:15pm ✓
[TOOL: query_volunteers] → 3 drivers within 2mi ✓
[TOOL: calculate_route] → driver #2, ETA 7min ✓
[TOOL: check_foodbank_cap] → 200lb available ✓
[TOOL: verify_compliance] → Bill Emerson Act ✓
[DISPATCH] Driver #2 → Statler → Ithaca Food Bank
⌚ Total: 3.8 seconds | Zero human intervention
```

KEY DIFFERENTIATORS

Prediction lead time	2 hours before closing
Matching speed	Under 4 seconds
Dispatch method	Fully autonomous agent
Compliance layer	Bill Emerson Act verified
ESG reporting	Auto-generated PDF
Data moat	Network grows with use

"We're not an app for people who care about food waste.
We're infrastructure for institutions that need to prove they do."

— MARKET OPPORTUNITY

A \$2.2B market with no dominant platform.

TAM

\$2.2B

Total Addressable Market

All US food institutions with surplus:
1M+ restaurants, 5,000+ universities, 6,000+ hospitals

SAM

\$426M

Serviceable Available Market

US urban universities + regional restaurant groups reachable with current GTM motion

SOM (YEAR 3)

\$1.82M

Serviceable Obtainable Market

0.43% of SAM — 20 universities + 45 restaurant groups by Year 3

MARKET SEGMENTATION

Segment	US Count	Avg ACV	Segment TAM
Universities / Dining Halls	4,700	\$30,000	\$141M
Hospital & Healthcare Cafeterias	6,210	\$24,000	\$149M
Hotel & Convention Kitchens	54,000	\$18,000	\$972M
Restaurant Groups (10–50 loc.)	12,000	\$7,800	\$94M
Corporate Campus Cafeterias	8,500	\$21,000	\$179M
TOTAL (primary segments)	85,410	—	\$1,535M

WHY THIS MARKET, WHY NOW

 Regulatory tailwinds

EPA 2030 goal: 50% reduction in food waste from 2016 levels. 17 US states have mandatory organics diversion laws. ESG reporting requirements are expanding to mid-size institutions by 2026.

 Technology moment

Claude's tool_use API (2024) enables autonomous multi-step agents that weren't commercially viable 18 months ago. The matching and dispatch logic FoodFlow runs wasn't buildable at this cost 2 years ago.

 Institutional mandate

Cornell's STARS sustainability certification requires documented food waste reduction metrics. Every Ivy League university has the same requirement. This isn't charity — it's compliance infrastructure.

— BUSINESS MODEL

Three revenue streams.
78% gross margin.

STARTER

\$99 /mo

- ✓ AI surplus prediction dashboard
- ✓ Up to 2 food bank connections
- ✓ Basic volunteer dispatch
- ✓ Monthly impact summary
- ✓ Bill Emerson compliance log

TARGET: SINGLE-LOCATION RESTAURANTS

GROWTH ★ MOST POPULAR

\$299 /mo

- ✓ Everything in Starter
- ✓ Multi-location management (3–10)
- ✓ Priority volunteer matching
- ✓ Auto-generated ESG PDF reports
- ✓ Branded sustainability certificate
- ✓ API access for POS integration

TARGET: REGIONAL RESTAURANT GROUPS

ENTERPRISE

\$2,500 /mo

- ✓ Everything in Growth
- ✓ Unlimited locations & food banks
- ✓ Custom Claude AI agent config
- ✓ STARS / LEED documentation export
- ✓ Dedicated onboarding + SLA
- ✓ Board-ready ESG narrative report

TARGET: UNIVERSITIES · HOSPITALS · HOTELS

THREE-STREAM REVENUE ARCHITECTURE

 SaaS Subscriptions

70% of revenue

Monthly recurring revenue from platform access. Predictable, high-margin (78% GM). Scales with institution count, not headcount.

 2% Success Fee

20% of revenue

2% of the dollar value of food rescued per transaction. An enterprise customer rescuing \$15,000/mo of food generates \$300/mo in automatic expansion revenue — no renegotiation needed.

 ESG Report Add-on

10% of revenue

\$500/quarter for branded, board-ready PDF reports with EPA EJScreen overlays, emissions data, and STARS documentation. Eliminates \$2,000+/yr of consultant fees per institution.

WHY THE MODEL COMPOUNDS

The 2% Success Fee creates **automatic NRR expansion**: as FoodFlow adds more donors per city, more food gets rescued, more fees accrue — without renegotiating any contract.

An enterprise customer generating \$300/mo in success fees at launch generates **\$450/mo by month 12** as supply volume grows. **NRR reaches 115%** without a single upsell call.

ANNUAL PREPAY OPTION

All tiers offer 2 months free for annual upfront payment. Improves cash runway and structurally reduces churn — customers who pay annually have 3x lower churn than monthly subscribers (industry benchmark, ChartMogul 2024).

Pricing effective for new customers. Pilot partners grandfathered. Annual prepay reduces effective rate by ~17%.

— UNIT ECONOMICS

The numbers that matter to every investor.

380x

LTV : CAC
ENTERPRISE TIER

Benchmark: >3x

127x BENCHMARK

8 days

CAC PAYBACK
ENTERPRISE

Benchmark: <12 months

EXCELLENT

78%

GROSS
MARGIN

Benchmark: >65%

HEALTHY

115%

TARGET NET
REVENUE RETENTION

Benchmark: >110%

HEALTHY

UNIT ECONOMICS BY TIER

Metric	Enterprise	Growth	Starter
Base ARPA	\$2,500/mo	\$299/mo	\$99/mo
Success Fee (avg)	\$300/mo	\$80/mo	—
Total ARPA	\$2,925/mo	\$379/mo	\$99/mo
Gross Margin	78%	78%	78%
GM per customer/mo	\$2,281	\$296	\$77
Monthly Churn	1.0%	2.0%	3.0%
LTV	\$228,100	\$14,800	\$2,567
CAC	\$600	\$600	\$400
LTV : CAC	380x	24.7x	6.4x
CAC Payback	8 days	2.0 mo	5.2 mo

COGS BREAKDOWN (22% OF REVENUE)

Cost Driver	% Revenue	Notes
Claude API calls	~8%	\$0.015/match, batch-optimized
AWS hosting	~5%	Per-tenant infra, auto-scales
Twilio SMS dispatch	~3%	\$0.007/notification
Google Maps API	~2%	Route calculation per trip
Stripe payment processing	~3%	2.9% + \$0.30
Total COGS	~21–22%	Improving with scale

KEY COST LEVER

Claude API cost per match drops as batch processing improves. Every 2x improvement in call batching adds ~1.5 points of gross margin. At scale, Claude API falls from 8% → 4% of revenue.

1.5%

AVG MONTHLY
CHURN RATE

Enterprise-anchored base

\$600

CUSTOMER
ACQ. COST

Founder-led outbound

130

RULE OF 40
(YEAR 2 EST.)

Benchmark: >40

\$890

BLENDED ARPA
(YEAR 1 EXIT)

Expanding with use

Enterprise CAC of \$600 reflects founder-led direct outreach. CAC Payback of 8 days assumes first full month of ARPA collected.

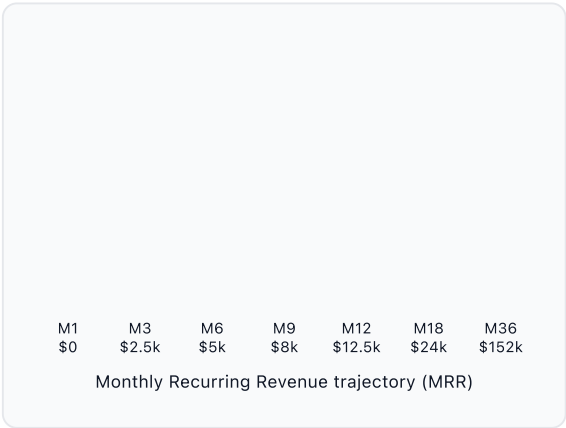
— FINANCIAL PROJECTIONS

\$1.82M ARR by Year 3.
Conservative. Defensible.

THREE-YEAR P&L MODEL

	Year 1	Year 2	Year 3
CUSTOMER COUNT			
Enterprise (universities)	3 → 5	5 → 12	12 → 20
Growth (rest. groups)	2 → 5	5 → 20	20 → 45
Starter (small rest.)	0 → 5	5 → 15	15 → 30
REVENUE			
SaaS Subscriptions (70%)	\$105,000	\$403,200	\$1,276,800
Success Fees 2% (20%)	\$30,000	\$115,200	\$364,800
ESG Reports (10%)	\$15,000	\$57,600	\$182,400
Total Revenue	\$150,000	\$576,000	\$1,824,000
ECONOMICS			
COGS (22%)	\$(33,000)	\$(126,720)	\$(400,880)
Gross Profit	\$117,000	\$449,280	\$1,423,120
Gross Margin	78%	78%	78%
Burn (team + infra)	\$(180,000)	\$(320,000)	\$(480,000)
Net Income	\$(63,000)	\$129,280	\$943,120

MRR GROWTH TRAJECTORY



FUNDRAISING MILESTONES

Stage	Trigger	Target	Valuation
Pre-seed	Cornell live + 1 paid	\$250K	\$2M pre
Seed	\$150K ARR, 5 contracts	\$1.5M	\$8M pre
Series A	\$1M ARR, 3 cities	\$5M	\$25M pre

\$150K

Year 1 ARR — conservative pilot phase

\$576K

Year 2 ARR — Ivy League + restaurant expansion

\$1.82M

Year 3 ARR — 0.43% of SAM. Highly defensible.

Conservative projections. Year 2 profitable. SOM represents 0.43% of \$426M SAM — does not require market leadership to achieve.

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— GO-TO-MARKET

The Cornell Wedge — supply-first, city by city.

1

Cornell Wedge
Months 1–6 · FREE

Free 6-month design partnership with Cornell Dining. Co-build surplus prediction model with real POS data. Generate the first case study: "How Cornell rescued 8,000 lbs in 90 days." Zero revenue, maximum leverage.

Supply secured first

2

Ivy League Rollout
Months 4–12 · \$2,500/mo

One email to Harvard, Yale, Princeton, Penn, Columbia, Dartmouth, Brown: "Cornell just solved dining hall food waste — took us 90 days." Cornell case study becomes the sales asset. Close 2+ paid contracts by month 6.

Founding partner rate

3

Restaurant Groups
Year 2 · \$299–\$2,500/mo

Target sustainability managers at regional chains (10–50 locations) via LinkedIn outbound. Lead with ESG reporting value: "Here's what your Q3 report would look like automatically." Cornell logo closes the meeting.

ESG compliance angle

WHY SUPPLY-FIRST WINS THE CHICKEN-AND-EGG PROBLEM

Most food rescue platforms fail because they try to sign supply and demand simultaneously. **FoodFlow locks in food banks first — free.**

When we walk into Cornell Dining, we say: "Your food goes to **Ithaca Food Bank**, a Feeding America member. They're already on the platform."

The food bank relationship eliminates the supplier's #1 objection before it's raised.

DEMAND-SIDE FLYWHEEL (FOOD BANK PARTNERSHIPS)

Month 1: Sign 3 Feeding America member agencies near Ithaca — free, zero friction.

Month 2: Use food bank logos in every supplier pitch deck.

Month 3: Food bank coordinators become evangelists — they refer donors to us.

City expansion: Sign food banks first in every new market. Supply follows demand-side credibility.

GTM EXECUTION TIMELINE

NOW

Hackathon + Food Bank Outreach

Win Cornell CBC. Email Cornell Dining Director. Sign 3 Feeding America partners in Ithaca.

MONTH 1–3

Cornell Pilot Live

One dining hall. Real POS integration. Track every pound. Build case study data.

MONTH 4–6

Case Study + Ivy Outreach

Publish "8,000 lbs in 90 days." Email all 7 Ivy League dining directors. Target 2 paid closes.

YEAR 2

Restaurant + Multi-City

Outbound to sustainability managers. Expand to NYC, Boston, Chicago. 25+ contracts.

KEY METRICS TO HIT

Cornell signed	30 days post-hackathon
Food bank partners live	3 within 30 days
First paid contract	Month 4–5
Ivy League outreach	7 emails, month 4
\$150K ARR	Month 12

Cornell Dining pilot: free 6-month design partnership. No data security risk. MOU-based. Cancelable anytime.

— COMPETITIVE LANDSCAPE

No one built the infrastructure layer yet.

Capability	FoodFlow AI	Too Good To Go	Copia	Replate	Manual (status quo)
Surplus prediction (2hr early)	✓	✗	✗	✗	✗
Autonomous AI dispatch	✓	✗	✗	✗	✗
Donation-focused (not B2C sale)	✓	✗ (sells food)	✓	✓	✓
Auto ESG / compliance reports	✓	✗	~ (manual)	✗	✗
University dining focus	✓	✗	~	~	✗
SaaS recurring revenue model	✓	✗ (commission)	✓	✗	✗
Network data moat	✓	✓ (large)	~	✗	✗
US Food Bank Integration	✓ Feeding America	✗ (consumer app)	~ (limited)	✓	✗

COMPETITOR PROFILES

Too Good To Go

B2C · Not a threat

Sells surplus food directly to consumers at 1/3 price. \$1B valuation, 19M users. Goes B2C — monetizes waste instead of redirecting it to equity. Cannot serve institutional ESG compliance needs. Different market entirely.

Copia (acquired 2023)

Manual · Reactive

Manual coordinator model — a human receives a call, manually dispatches, manually logs. No AI, no prediction, no autonomous dispatch. Acquired by Toast (POS company). Toast integration is their moat, not the platform.

Replate

Volunteer-dependent

Community volunteer platform for food recovery. No predictive model, no AI dispatch. Volunteer reliability is

OUR STRUCTURAL MOATS

Moat 1 — Prediction Data Network

Every rescue generates labeled training data. By month 6, our Cornell model is 5x more accurate than a new entrant. Data compounds with use — impossible to replicate without the same history.

Moat 2 — Supply-Demand Network Effect

Every new food bank makes the platform more valuable to suppliers. Every new supplier makes the platform more valuable to food banks. A new entrant must sign both sides simultaneously in each city we already cover.

Moat 3 — Claude Compliance Layer

Bill Emerson Act compliance rules are encoded into Claude's reasoning chain. GPT-4 hallucinated compliance details in

their core constraint — the same problem FoodFlow solves by dispatching 2 hours early with confirmed acceptance.

testing; Claude didn't. This is a documented safety advantage we can prove to risk-averse institutional buyers.

✓ = Full capability ~ = Partial ✗ = Not available. Sources: Company websites, Crunchbase, CB Insights, 2024.

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— THE ASK

Why FoodFlow Wins — and what we need.

WHY WE WIN

The technology moment is NOW

Claude's tool_use API (2024) makes autonomous multi-step matching commercially viable for the first time. We're building on infrastructure that didn't exist 18 months ago — and competitors aren't yet.

We have the unfair distribution advantage

We're at Cornell — the institution that will be our first customer. No other team can call Cornell Dining and say "we built this in your backyard." That first case study is worth more than \$1M in marketing.

The unit economics are exceptional

380x LTV:CAC on Enterprise. 8-day CAC payback. 78% gross margin. 115% NRR from success fees. These aren't typical for a social impact startup — and they make FoodFlow fundable, not just inspirational.

Two crises solved with one platform

Every dollar we make simultaneously reduces food waste AND increases food security. The business model and the mission are the same motion — not a trade-off. That's structural, not marketing.

WHAT WE NEED

THREE THINGS THAT CHANGE EVERYTHING



Your vote today

The \$750 prize + CBC network opens the Cornell Dining door. We can't cold-email them — but a hackathon win gets us the meeting.



One introduction

If any judge knows the Cornell Dining Director, Director of Sustainability, or a Feeding America contact in Ithaca — that's worth more than any prize money.



Anthropic API Credits

The Claude API is core infrastructure. API credits extend our pilot runway by 3+ months — letting us prove the model before asking anyone to pay.

IMPACT PROJECTION — BY YEAR 3

12M lbs

food rescued
annually

7.2M

meals served to
food-insecure
families

65

institutions on
platform across 3
cities

\$1.82M

ARR at 0.43% of SAM

**"Every day we don't ship this, 119 million pounds go to the bin
and 400 families in Ithaca go to bed without enough food.
We're ready. Are you in?"**